Biodiversity and Conservation

If an alien from a distant galaxy were to visit our planet Earth, the first thing that would amaze and baffle him would most probably be the enormous diversity of life that he would encounter.

The scale of it

- even for humans, the rich variety of living organisms with which they share this planet never ceases to astonish and fascinate
- the common man would find it hard to believe that there are more than 20,000 species of ants, 3,00,000 species of beetles, 28,000 species of fishes and nearly 20,000 species of orchids

The questions ecologists ask of it

- why there are so many species, and whether such great diversity existed throughout earth's history
- how this diversification came about
- how and why this diversity is important to the biosphere, and whether it would function any differently if the diversity were much less
- how humans benefit from the diversity of life

What biodiversity means

- in our biosphere immense diversity, or heterogeneity, exists not only at the species level but at all levels of biological organisation, ranging from macromolecules within cells to biomes
- biodiversity is the term popularised by the sociobiologist Edward Wilson to describe the combined diversity at all the levels of biological organisation

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13.1 Levels of biodiversity

Genetic diversity

- a single species might show high diversity at the genetic level over its distributional range
- the genetic variation shown by the medicinal plant Rauwolfia vomitoria growing in different Himalayan ranges might be in terms of the potency and concentration of the active chemical, reserpine, that the plant produces
- India has more than 50,000 genetically different strains of rice, and 1,000 varieties of mango

Species and ecological diversity

- species diversity is the diversity at the species level, as when the Western Ghats have a greater amphibian species diversity than the Eastern Ghats
- ecological diversity is diversity at the ecosystem level, as when India, with its deserts, rain forests, mangroves, coral reefs, wetlands, estuaries and alpine meadows, has a greater ecosystem diversity than a Scandinavian country such as Norway
- it has taken millions of years of evolution to accumulate this rich diversity in nature, but all that wealth could be lost in less than two centuries if the present rates of species loss continue, so biodiversity and its conservation are now vital environmental issues of international concern

13.2 How many species there are

What is recorded, and what is guessed

- since there are published records of all the species discovered and named, we know how many have been recorded so far, but how many species there are on earth is not easy to answer
- according to the International Union for Conservation of Nature and Natural Resources (IUCN) in 2004, the total number of plant and animal species described so far is slightly more than 1.5 million, and we have no clear idea of how many are yet to be discovered and described
- estimates vary widely and many of them are only educated guesses, and for many taxonomic groups species inventories are more complete in temperate than in tropical countries

How the estimate is made

- an overwhelmingly large proportion of the species waiting to be discovered are in the tropics
- so biologists make a statistical comparison of the temperate to tropical species richness of an exhaustively studied group of insects, and extrapolate that ratio to other groups of animals and plants, to reach a gross estimate of the total number of species on earth
- some extreme estimates range from 20 to 50 million, but a more conservative and scientifically sound estimate made by Robert May places the global species diversity at about 7 million

Animals, more than 70 per cent of all species recorded

Insects, more than 70 per cent of the animals

Plants, no more than 22 per cent of the total

Fungi, outnumbering fishes, amphibians, reptiles and mammals combined

Figure 13.A Representing global biodiversity: proportionate number of species of major taxa of plants, invertebrates and vertebrates

13.3 What the inventories show

The proportions

- more than 70 per cent of all the species recorded are animals
- plants, including algae, fungi, bryophytes, gymnosperms and angiosperms, comprise no more than 22 per cent of the total
- among animals, insects are the most species-rich taxonomic group, making up more than 70 per cent of the total, so that out of every 10 animals on this planet, 7 are insects, which leaves the enormous diversification of insects itself to be explained
- the number of fungi species in the world is more than the combined total of the species of fishes, amphibians, reptiles and mammals

What the estimates leave out

- these estimates give no figures for prokaryotes, and biologists are not sure how many prokaryotic species there might be
- the problem is that conventional taxonomic methods are not suitable for identifying microbial species, and many species are simply not culturable under laboratory conditions
- if biochemical or molecular criteria are accepted for delineating species in this group, then their diversity alone might run into millions

13.4 India's share

A mega diversity country

- although India has only 2.4 per cent of the world's land area, its share of the global species diversity is an impressive 8.1 per cent, which makes it one of the 12 mega diversity countries of the world
- nearly 45,000 species of plants and twice as many of animals have been recorded from India
- if May's global estimates are accepted, only 22 per cent of the total species have been recorded so far, and applying that proportion to India's figures gives probably more than 1,00,000 plant species and more than 3,00,000 animal species yet to be discovered and described
- whether the inventory of the biological wealth of the country can ever be completed depends on the immense trained manpower, the taxonomists, and the time required, and the situation looks more hopeless still because a large fraction of these species faces the threat of extinction even before they are discovered

13.5 Latitudinal gradients

Diversity falls towards the poles

- the diversity of plants and animals is not uniform throughout the world but shows a rather uneven distribution, the most well-known pattern being the latitudinal gradient
- in general species diversity decreases as we move away from the equator towards the poles, and with very few exceptions the tropics, a latitudinal range of 23.5° N to 23.5° S, harbour more species than temperate or polar areas
- Colombia, located near the equator, has nearly 1,400 species of birds, while New York at 41° N has 105 species and Greenland at 71° N only 56, and India, with much of its land area in the tropical latitudes, has more than 1,200 species of birds

The tropical forest

- a forest in a tropical region such as Equador has up to 10 times as many species of vascular plants as a forest of equal area in a temperate region such as the Midwest of the USA
- the largely tropical Amazonian rain forest in South America has the greatest biodiversity on earth, being home to more than 40,000 species of plants, 3,000 of fishes, 1,300 of birds, 427 of mammals, 427 of amphibians, 378 of reptiles and more than 1,25,000 invertebrates
- scientists estimate that in these rain forests there might be at least two million insect species waiting to be discovered and named

Species richness rising with area explored

The curve flattening, a rectangular hyperbola

On a log scale the same relationship becoming a straight line

Z, the slope of that line

Figure 13.B Showing species area relationship. Note that on log scale the relationship becomes linear

13.6 Why the tropics are richer

Three hypotheses

- speciation is generally a function of time, and unlike temperate regions subjected to frequent glaciations, tropical latitudes have remained relatively undisturbed for millions of years and so had a long evolutionary time for species diversification
- tropical environments, unlike temperate ones, are less seasonal, more constant and predictable, and such environments promote niche specialisation and greater species diversity
- there is more solar energy available in the tropics, which contributes to higher productivity, and this in turn might contribute indirectly to greater diversity

13.7 Species-area relationships

The curve, and the equation

- during his pioneering explorations in South American jungles, the great German naturalist and geographer Alexander von Humboldt observed that within a region species richness increased with increasing explored area, but only up to a limit
- the relation between species richness and area for a wide variety of taxa, angiosperm plants, birds, bats and freshwater fishes, turns out to be a rectangular hyperbola
- on a logarithmic scale the relationship is a straight line described by the equation log S = log C + Z log A, where S is species richness, A is area, Z is the slope of the line, that is the regression coefficient, and C is the Y-intercept

What Z turns out to be

- the value of Z lies in the range of 0.1 to 0.2 regardless of the taxonomic group or the region, so whether it is plants in Britain, birds in California or molluscs in New York state, the slopes of the regression line are amazingly similar
- but among very large areas such as entire continents the slope of the line is much steeper, with Z values in the range of 0.6 to 1.2
- for frugivorous, that is fruit-eating, birds and mammals in the tropical forests of different continents the slope is found to be 1.15, which leaves the meaning of steeper slopes to be worked out

13.8 Species diversity and the ecosystem

An unsettled question

- whether the number of species in a community really matters to the functioning of the ecosystem is a question ecologists cannot answer definitively
- for many decades ecologists believed that communities with more species tend to be more stable than those with fewer

13.9 Stability, and the rivet popper

What stability means

- a stable community should not show too much variation in productivity from year to year, it must be either resistant or resilient to occasional disturbances, natural or man-made, and it must also be resistant to invasions by alien species
- how these attributes are linked to species richness in a community is not known, but David Tilman's long-term ecosystem experiments using outdoor plots provide some tentative answers
- Tilman found that plots with more species showed less year-to-year variation in total biomass, and also showed that increased diversity contributed to higher productivity
- we may not understand completely how species richness contributes to the well-being of an ecosystem, yet we know enough to realise that rich biodiversity is not only essential for ecosystem health but imperative for the very survival of the human race on this planet

The rivet popper hypothesis

- at a time when species are being lost at an alarming pace, one might ask whether it really matters if a few species become extinct, whether Western Ghats ecosystems would be less functional if one of their tree frog species were lost forever, and how quality of life is affected if there were only 15,000 rather than 20,000 species of ants on earth
- there are no direct answers to such näive questions, but a proper perspective can be developed through an analogy used by the Stanford ecologist Paul Ehrlich
- in an airplane, the ecosystem, all parts are joined together using thousands of rivets, the species, and if every passenger travelling in it starts popping a rivet to take home, that is causing a species to become extinct, it may not affect flight safety initially, but as more and more rivets are removed the plane becomes dangerously weak over a period of time
- which rivet is removed may also be critical, since loss of rivets on the wings, that is key species that drive major ecosystem functions, is obviously a more serious threat to flight safety than loss of a few rivets on the seats or windows inside the plane

13.10 Loss of biodiversity

What has already gone

- while it is doubtful whether any new species are being added through speciation into the earth's treasury of species, there is no doubt about their continuing losses, and the biological wealth of our planet has been declining rapidly, with the accusing finger clearly pointing to human activities
- the colonisation of tropical Pacific Islands by humans is said to have led to the extinction of more than 2,000 species of native birds
- the IUCN Red List of 2004 documents the extinction of 784 species, including 338 vertebrates, 359 invertebrates and 87 plants, in the last 500 years
- some examples of recent extinctions include the dodo of Mauritius, the quagga of Africa, the thylacine of Australia, Steller's Sea Cow of Russia and three subspecies, Bali, Javan and Caspian, of tiger, and the last twenty years alone have witnessed the disappearance of 27 species

13.11 The sixth extinction

The threat now

- careful analysis of records shows that extinctions across taxa are not random, and some groups such as amphibians appear to be more vulnerable
- more than 15,500 species world-wide are facing the threat of extinction
- presently 12 per cent of all bird species, 23 per cent of all mammal species, 32 per cent of all amphibian species and 31 per cent of all gymnosperm species in the world face that threat

How this episode differs

- a study of the history of life on earth through fossil records shows that large-scale loss of species like the one now being witnessed has happened earlier too, even before humans appeared on the scene
- during the long period, more than 3 billion years, since the origin and diversification of life on earth there were five episodes of mass extinction of species
- the difference lies in the rates, the current species extinction rates being estimated at 100 to 1,000 times faster than in pre-human times, with our activities responsible for that speed
- ecologists warn that if the present trends continue, nearly half of all the species on earth might be wiped out within the next 100 years

What loss of biodiversity leads to

- decline in plant production
- lowered resistance to environmental perturbations such as drought
- increased variability in certain ecosystem processes such as plant productivity, water use, and pest and disease cycles

13.12 Causes of biodiversity loss

Habitat loss and fragmentation

- the accelerated rates of species extinction now faced are largely due to human activities, and there are four major causes, described by the sobriquet the Evil Quartet
- habitat loss and fragmentation is the most important cause driving animals and plants to extinction, and the most dramatic examples come from tropical rain forests, which once covered more than 14 per cent of the earth's land surface and now cover no more than 6 per cent
- they are being destroyed fast, so that in the time it takes to read this chapter 1000 more hectares of rain forest would have been lost, and the Amazon rain forest, so huge that it is called the lungs of the planet and harbouring probably millions of species, is being cut and cleared for cultivating soya beans or for conversion to grasslands for raising beef cattle
- besides total loss, the degradation of many habitats by pollution also threatens the survival of many species, and when large habitats are broken up into small fragments by human activities, mammals and birds requiring large territories and certain animals with migratory habits are badly affected, leading to population declines

13.13 The rest of the evil quartet

Over-exploitation

- humans have always depended on nature for food and shelter, but when need turns to greed it leads to over-exploitation of natural resources
- many species extinctions in the last 500 years, such as Steller's sea cow and the passenger pigeon, were due to overexploitation by humans
- presently many marine fish populations around the world are over harvested, endangering the continued existence of some commercially important species

Alien species invasions

- when alien species are introduced unintentionally or deliberately, for whatever purpose, some of them turn invasive and cause the decline or extinction of indigenous species
- the Nile perch introduced into Lake Victoria in east Africa led eventually to the extinction of an ecologically unique assemblage of more than 200 species of cichlid fish in the lake
- invasive weed species such as carrot grass (Parthenium), Lantana and water hyacinth (Eicchornia) cause environmental damage and pose a threat to our native species
- the recent illegal introduction of the African catfish Clarias gariepinus for aquaculture purposes is posing a threat to the indigenous catfishes in our rivers

Co-extinctions

- when a species becomes extinct, the plant and animal species associated with it in an obligatory way also become extinct
- when a host fish species becomes extinct, its unique assemblage of parasites meets the same fate
- another example is a coevolved plant-pollinator mutualism, where the extinction of one invariably leads to the extinction of the other

13.14 Why we should conserve biodiversity

The narrowly utilitarian argument

- there are many reasons, some obvious and others not so obvious, but all equally important, and they group into three categories, narrowly utilitarian, broadly utilitarian and ethical
- humans derive countless direct economic benefits from nature, that is food such as cereals, pulses and fruits, firewood, fibre, construction material, industrial products such as tannins, lubricants, dyes, resins and perfumes, and products of medicinal importance
- more than 25 per cent of the drugs currently sold in the market worldwide are derived from plants, and 25,000 species of plants contribute to the traditional medicines used by native peoples around the world, and nobody knows how many more medicinally useful plants there are in tropical rain forests waiting to be explored
- with increasing resources put into bioprospecting, that is exploring molecular, genetic and species-level diversity for products of economic importance, nations endowed with rich biodiversity can expect to reap enormous benefits

13.15 Broadly utilitarian and ethical arguments

The broadly utilitarian argument

- biodiversity plays a major role in many ecosystem services that nature provides
- the fast-dwindling Amazon forest is estimated to produce, through photosynthesis, 20 per cent of the total oxygen in the earth's atmosphere, and an idea of the economic value of that service can be got from what a neighborhood hospital spends on a cylinder of oxygen
- pollination, without which plants cannot give us fruits or seeds, is another service, provided through the pollinators layer of bees, bumblebees, birds and bats, which leaves the cost of accomplishing pollination without natural pollinators to be reckoned
- there are other intangible benefits we derive from nature, the aesthetic pleasures of walking through thick woods, watching spring flowers in full bloom or waking up to a bulbul's song in the morning, and whether a price tag can be put on such things is itself a question

The ethical argument

- it relates to what we owe to the millions of plant, animal and microbe species with whom we share this planet
- philosophically or spiritually, we need to realise that every species has an intrinsic value, even if it may not be of current or any economic value to us
- we have a moral duty to care for their well-being and to pass on our biological legacy in good order to future generations

13.16 How we conserve biodiversity

In situ and ex situ

- when we conserve and protect the whole ecosystem, its biodiversity at all levels is protected, so we save the entire forest to save the tiger, and this approach is called in situ, that is on site, conservation
- where an animal or plant is endangered or threatened, that is facing a very high risk of extinction in the wild in the near future, and needs urgent measures to save it, ex situ, that is off site, conservation is the desirable approach

Biodiversity hotspots

- faced with the conflict between development and conservation, many nations find it unrealistic and economically not feasible to conserve all their biological wealth, and invariably the number of species waiting to be saved far exceeds the conservation resources available
- eminent conservationists have addressed this on a global basis by identifying for maximum protection certain biodiversity hotspots, regions with very high levels of species richness and a high degree of endemism, that is species confined to that region and found nowhere else
- initially 25 hotspots were identified, and nine more have since been added, bringing the total in the world to 34, and these are also regions of accelerated habitat loss
- three of them, Western Ghats and Sri Lanka, Indo-Burma and Himalaya, cover our country's exceptionally high biodiversity regions, and although all the hotspots together cover less than 2 per cent of the earth's land area, the number of species they collectively harbour is extremely high

13.17 In situ conservation in India

Reserves, parks and sacred groves

- strict protection of the hotspots could reduce the ongoing mass extinctions by almost 30 per cent
- in India, ecologically unique and biodiversity-rich regions are legally protected as biosphere reserves, national parks and sanctuaries, and India now has 14 biosphere reserves, 90 national parks and 448 wildlife sanctuaries
- India also has a history of religious and cultural traditions that emphasised protection of nature, and in many cultures tracts of forest were set aside, with all the trees and wildlife within venerated and given total protection
- such sacred groves are found in the Khasi and Jaintia Hills in Meghalaya, the Aravalli Hills of Rajasthan, the Western Ghat regions of Karnataka and Maharashtra and the Sarguja, Chanda and Bastar areas of Madhya Pradesh, and in Meghalaya the sacred groves are the last refuges for a large number of rare and threatened plants

13.18 Ex situ conservation

Out of the habitat

- in this approach threatened animals and plants are taken out of their natural habitat and placed in a special setting where they can be protected and given special care, and zoological parks, botanical gardens and wildlife safari parks serve this purpose
- there are many animals that have become extinct in the wild but continue to be maintained in zoological parks
- in recent years the approach has advanced beyond keeping threatened species in enclosures, so that gametes of threatened species can now be preserved in viable and fertile condition for long periods using cryopreservation techniques, eggs can be fertilised in vitro, and plants can be propagated using tissue culture methods
- seeds of different genetic strains of commercially important plants can be kept for long periods in seed banks

A collective responsibility

- biodiversity knows no political boundaries, so its conservation is a collective responsibility of all nations
- the historic Convention on Biological Diversity, the Earth Summit, held in Rio de Janeiro in 1992, called upon all nations to take appropriate measures for the conservation of biodiversity and the sustainable utilisation of its benefits
- in a follow-up, at the World Summit on Sustainable Development held in 2002 in Johannesburg, South Africa, 190 countries pledged their commitment to achieve by 2010 a significant reduction in the current rate of biodiversity loss at global, regional and local levels

13.19 Summary

What biodiversity is

- since life originated on earth nearly 3.8 billion years ago there has been enormous diversification of life forms
- biodiversity refers to the sum total of diversity that exists at all levels of biological organisation, of particular importance being the diversity at genetic, species and ecosystem levels, and conservation efforts are aimed at protecting diversity at all these levels
- more than 1.5 million species have been recorded in the world, but there might still be nearly 6 million species on earth waiting to be discovered and named

13.20 Summary, continued

Numbers and patterns

- of the named species, more than 70 per cent are animals, of which 70 per cent are insects, and the group Fungi has more species than all the vertebrate species combined, while India, with about 45,000 species of plants and twice as many species of animals, is one of the 12 mega diversity countries of the world
- species diversity on earth is not uniformly distributed but shows interesting patterns, being generally highest in the tropics and decreasing towards the poles, the important explanations for tropical species richness being that the tropics had more evolutionary time, that they provide a relatively constant environment, and that they receive more solar energy, which contributes to greater productivity
- species richness is also a function of the area of a region, the species-area relationship being generally a rectangular hyperbolic function

Losses, and why they matter

- it is believed that communities with high diversity tend to be less variable, more productive and more resistant to biological invasions
- earth's fossil history reveals incidence of mass extinctions in the past, but the present rates of extinction, largely attributed to human activities, are 100 to 1000 times higher, nearly 700 species having become extinct in recent times and more than 15,500 species, of which more than 650 are from India, currently facing the threat of extinction
- the causes of high extinction rates at present include habitat loss and fragmentation, particularly of forests, over-exploitation, biological invasions and co-extinctions
- earth's rich biodiversity is vital for the very survival of mankind, and the reasons for conserving it are narrowly utilitarian, broadly utilitarian and ethical, since besides the direct benefits of food, fibre, firewood and pharmaceuticals there are many indirect benefits received through ecosystem services such as pollination, pest control, climate moderation and flood control, and we also have a moral responsibility to take good care of earth's biodiversity and pass it on in good order to our next generation

The two approaches

- biodiversity conservation may be in situ as well as ex situ
- in in situ conservation the endangered species are protected in their natural habitat, so that the entire ecosystem is protected, and recently 34 biodiversity hotspots in the world have been proposed for intensive conservation efforts, of which three, Western Ghats and Sri Lanka, Himalaya and Indo-Burma, cover India's rich biodiversity regions
- our country's in situ conservation efforts are reflected in its 14 biosphere reserves, 90 national parks, more than 450 wildlife sanctuaries and many sacred groves
- ex situ conservation methods include protective maintenance of threatened species in zoological parks and botanical gardens, in vitro fertilisation, tissue culture propagation and cryopreservation of gametes

13.21 Exercises

From this chapter

1 Name the three important components of biodiversity.
2 How do ecologists estimate the total number of species present in the world?

13.22 Exercises, continued

From this chapter, continued

1. Give three hypotheses for explaining why tropics show greatest levels of species richness.
2. What is the significance of the slope of regression in a species – area relationship?
3. What are the major causes of species losses in a geographical region?
4. How is biodiversity important for ecosystem functioning?
5. What are sacred groves? What is their role in conservation?
6. Among the ecosystem services are control of floods and soil erosion. How is this achieved by the biotic components of the ecosystem?
7. The species diversity of plants (22 per cent) is much less than that of animals (72 per cent). What could be the explanations to how animals achieved greater diversification?
8. Can you think of a situation where we deliberately want to make a species extinct? How would you justify it?

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